

Task Coordinates for Few-Shot Diffusion Adaptation: Results Against the Experimental Plan

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Abstract

Meta-Diffusion adapts a frozen denoising diffusion probabilistic model to a new data domain by fitting a low-dimensional task coordinate rather than by updating network weights. This report gives the results of the current experimental stage, block by block against the governing plan. The plan fixes the criterion in advance: the main claim holds only if the correct source task together with a learned source-to-target relation outperforms *both* relation-only transfer *and* target-only few-shot modelling. That criterion is met at every target budget measured, but the margin over target-only modelling narrows from +0.0167 at one target image to +0.0044 at twenty, so the method earns its keep chiefly where target data are scarce. The architectural claim does not survive: against matched FiLM conditioning the additive basis is indistinguishable on that same metric, and what remains is an efficiency result — the same generation quality from fewer than half the parameters. Seven of the ten deliverables are complete; three are not, and each is named with its reason.

1 CRITERION AND NOTATION

Criterion. The main claim is supported only if the correct source task *plus* the learned source-to-target relation outperforms *both* relation-only transfer *and* target-only few-shot modelling, in the low-target-data regime.

Notation. A set encoder maps a set of images to a task coordinate $z \in \mathbb{R}^k$. Writing z_S for the coordinate encoded from source-domain data and c for a descriptor of the source-to-target relation, the transport map predicts the target coordinate $\tilde{z}_T = z_S + \Delta_\gamma(z_S, c)$, taking no target image as input. M_S is the number of source images the encoder sees; K_T is the number of target images available at adaptation time. **FiLM**, feature-wise linear modulation, is the standard alternative to an additive basis: instead of adding a learned direction to the score it scales and shifts intermediate feature maps as a function of the coordinate. It is the baseline A2 measures against.

Statistics. Every comparison is a paired difference over episodes with a 95% confidence interval, and an asterisk marks an interval excluding zero.

The instruments. Unless a section says otherwise, a distance is the sliced Wasserstein distance, which the plan names as headline. Six measurements appear in this report; they are set out here once because A2 uses all of them and they do not agree.

- **Sliced Wasserstein distance**, pixel space. Both sets of images are projected onto many random directions; each projection is sorted and compared quantile by quantile, and the result is the root mean square over directions. Raw pixels, nothing learned. Lower is better.
- **Transformation statistic.** Three fixed convolutions per image: mean absolute Laplacian response, which is high-frequency energy; per-image standard deviation, which is dynamic range; and mean absolute residual above a local box average. Blur and contrast reduction both lower the first, only contrast lowers the second, so the triple separates them. No training and no weights, so nothing in it drifts between runs. It was built for CIFAR’s corruptions and reads nothing outside them. Lower is better.

- **Energy maximum mean discrepancy**, pixel space: $2\mathbb{E}\|a - b\| - \mathbb{E}\|a - a'\| - \mathbb{E}\|b - b'\|$ over mean pairwise Euclidean distances. It requires no kernel width, which is why it sits beside the Wasserstein distance rather than a Gaussian-kernel discrepancy. Lower is better.
- **Class consistency**. The share of an episode’s generated samples that a trained twenty-way classifier assigns to that episode’s own class. *Higher* is better. The classifier reports its own held-out accuracy beside every verdict; in A2 that ceiling is 38.9% on real images, so it is a weak instrument and is read as one.
- **Superclass consistency**. The same at the level of CIFAR-100’s twenty coarse classes. *Higher* is better.
- **Kernel Inception distance**. The unbiased squared maximum mean discrepancy under the cubic polynomial kernel $k(x, y) = (x \cdot y / d + 1)^3$, averaged over random subsets. **The features are not Inception features**: this project has no InceptionV3, so they come from the same small convolutional instrument quoted above. The estimator is standard and the feature space is local, so the number compares the arms measured here and cannot be set beside any published value. Lower is better.

Configuration. One checkpoint of the three-transformation CIFAR-100 domain-shift family — Gaussian blur, additive Gaussian noise, contrast reduction at severity 3 — a U-Net backbone of 128 base channels, coordinate dimension $k = 16$, 50,000 training steps read with exponentially moving averaged weights. Adaptation uses 25 refinement steps at rate 10^{-2} . Generation uses the deterministic implicit sampler, 50 reverse steps, 96 samples per cell, 60 episodes per cell.

2 STAGE 0 — PRECONDITIONS

	Precondition	Status
0.1	Integrate the three branches	Done. All three files coexist in the tree that produced A2.
0.2	Verify the split file	Done; hash matches the fixed value.
0.3	FiLM checkpoint	Done. Both A2 arms were retrained from scratch rather than recovered.
0.4	Evaluation primitives	Done. The kernel Inception distance, the last of the five, was written for A2.
0.5	Output directory	Done.

Table 1: Stage 0. The five preconditions the plan places before all other work, and whether each holds in the tree that produced the results below.

3 STAGE A — CIFAR-100, GENERATION-LEVEL EVIDENCE

3.1 A1 — Source-dependence controls

180 episodes, twenty classes by three transformations by $K_T \in \{1, 5, 20\}$, nine conditions.

Two arms are reported. *Transport* applies the map and stops, consulting no target image; it is the cleaner comparison against the source controls, because it asks what the source coordinate is worth with target evidence held out of the question. *Transport + refine* then refines the predicted coordinate on the K_T target images and is what the strategy table names as the method; the plan’s criterion is about it.

Conclusion. Both arms beat all three source controls at every budget, so the source coordinate is worth having however the method is deployed. The criterion also requires beating target-only modelling, and there the two arms part company at the largest budget: **the method with refinement keeps a margin at every budget measured, +0.0167, +0.0138 and +0.0044, each interval clear of zero**, while the unrefined arm has given that margin up by twenty target images. The criterion is therefore met across the range, but the margin at twenty target images is a quarter of the margin at one, so the honest statement remains that **source information is worth most when target evidence is scarce** and that its advantage narrows steadily as target evidence accumulates.

Arm, against	$K_T = 1$	$K_T = 5$	$K_T = 20$
<i>Transport</i> — no target image consulted			
Mean z_S	+0.0301 $\pm$ 0.0072*	+0.0346 $\pm$ 0.0085*	+0.0338 $\pm$ 0.0080*
Shuffled z_S	+0.0416 $\pm$ 0.0125*	+0.0453 $\pm$ 0.0149*	+0.0436 $\pm$ 0.0144*
Relation-only	+0.0254 $\pm$ 0.0078*	+0.0285 $\pm$ 0.0088*	+0.0295 $\pm$ 0.0087*
Target-only	+0.0181 $\pm$ 0.0085*	+0.0114 $\pm$ 0.0072*	−0.0028 $\pm$ 0.0073
Behind the oracle	+0.0246 $\pm$ 0.0085*	+0.0288 $\pm$ 0.0095*	+0.0244 $\pm$ 0.0099*
<i>Transport + refine</i> — the method			
Mean z_S	+0.0286 $\pm$ 0.0074*	+0.0369 $\pm$ 0.0090*	+0.0410 $\pm$ 0.0112*
Shuffled z_S	+0.0402 $\pm$ 0.0125*	+0.0476 $\pm$ 0.0158*	+0.0508 $\pm$ 0.0161*
Relation-only	+0.0240 $\pm$ 0.0079*	+0.0309 $\pm$ 0.0086*	+0.0367 $\pm$ 0.0101*
Target-only	+0.0167 $\pm$ 0.0086*	+0.0138 $\pm$ 0.0055*	+0.0044 $\pm$ 0.0031*
Behind the oracle	+0.0260 $\pm$ 0.0085*	+0.0265 $\pm$ 0.0091*	+0.0172 $\pm$ 0.0085*

Table 2: A1. Paired differences over 180 episodes; positive favours the arm named, except in the last line of each block, where positive means the arm is behind the oracle. The refined arm is re-summarised from the run’s own per-episode dump, not regenerated.

Against the source controls the refined arm’s margins *grow* with K_T — against the mean source coordinate, from +0.0286 to +0.0410 — which is what a procedure that consumes target images should do against controls holding a coordinate that is simply wrong. A gap to the oracle remains at every budget and for both arms.

3.2 A2 — Additive basis against FiLM

Two networks trained from scratch under identical conditions, evaluated on the same 180 held-out episodes with one shared initial noise per episode. The arms share the backbone size, the set encoder, the transport map, k , the base head and the loss; only the conditioning differs. In the basis arm the coordinate never enters the backbone, weighting score-perturbation directions added to the output, $\hat{\varepsilon}_z = \hat{\varepsilon}_0 + \sum_l z_l R_l(x_t, t)$. In the FiLM arm it is fed to the backbone and folded into every residual block’s scale and shift. The parameters carrying the coordinate are **55,344 for the basis against 118,784 for FiLM**; FiLM is deliberately the larger, since a stronger baseline is the one worth beating.

Instrument	Transport arm	Oracle arm
Sliced Wasserstein (<i>headline</i>)	tie at all three budgets	basis , all three
Transformation statistic	FiLM , all three	tie at all three
Energy discrepancy	FiLM , two of three	tie at all three
Class consistency	FiLM , two of three	basis , all three
Superclass consistency	FiLM , all three	tie at all three
Kernel Inception distance	basis (not paired)	tie

Table 3: A2, the verdict by instrument. A cell reads *tie* when the 95% interval contains zero at that budget. The transport arm is the method’s own; the oracle arm refines on abundant target data and bounds what each conditioning mechanism can reach.

Sliced Wasserstein, FiLM minus basis	$K_T = 1$	$K_T = 5$	$K_T = 20$
transport	+0.0002 $\pm$ 0.0070	−0.0020 $\pm$ 0.0076	−0.0055 $\pm$ 0.0074
transport + refine	−0.0008 $\pm$ 0.0074	−0.0014 $\pm$ 0.0086	−0.0052 $\pm$ 0.0089
reuse z_S	−0.0191 $\pm$ 0.0072*	−0.0050 $\pm$ 0.0093	−0.0071 $\pm$ 0.0101
oracle	+0.0163 $\pm$ 0.0105*	+0.0183 $\pm$ 0.0118*	+0.0147 $\pm$ 0.0119*

Table 4: A2 on the headline metric, signed so that positive favours the basis. The other instruments are summarised in the table above; their values are in `A2_basis_vs_film.txt`.

The oracle gap, and the explanation that failed. The basis leads at the oracle. The natural reading is that it offers a loss surface refinement can descend further, and that transport does not land close enough to exploit it. A probe tested that from the same start, on the same data, under the same budget and noise draws, and reversed it: the basis descends *less* than FiLM from the transported coordinate ($-0.0024 \pm 0.0005^*$ as a fraction of each arm’s own starting loss) and much less from $z = 0$ ($-0.0796 \pm 0.0121^*$), along a segment that is rougher rather than smoother. The probe carried its own rule for that outcome: the gap then needs another explanation and must be reported without this one. None is offered here.

Conclusion. On the headline metric the additive reverse-dynamics basis and matched FiLM conditioning are indistinguishable on the method’s own arm — the interval contains zero at every target budget, with refinement and without. Four of the five paired instruments place FiLM *ahead* there, three significantly. The basis leads only at the oracle, and the mechanism proposed for that lead was contradicted. What survives is an efficiency claim rather than a performance one, and it should be stated as one: **the basis reaches the same generation quality as FiLM while carrying the task coordinate in 55,344 parameters against 118,784, fewer than half.** That is a real property of the architecture, and it is not the claim the stage set out to establish.

3.3 A3 — Target-scarcity sweep

300 episodes at $K_T \in \{1, 2, 5, 10, 20\}$, six strategies.

Strategy	$K_T = 1$	$K_T = 2$	$K_T = 5$	$K_T = 10$	$K_T = 20$
No adaptation	$1.616 \pm .094$	$1.620 \pm .091$	$1.609 \pm .095$	$1.621 \pm .094$	$1.586 \pm .090$
Reuse z_S	$1.439 \pm .106$	$1.212 \pm .098$	$0.825 \pm .086$	$0.630 \pm .058$	$0.521 \pm .050$
Transport	$0.299 \pm .045$	$0.297 \pm .045$	$0.299 \pm .042$	$0.313 \pm .041$	$0.293 \pm .040$
Transport + refine	$0.291 \pm .045$	$0.294 \pm .047$	$0.295 \pm .036$	$0.312 \pm .043$	$0.322 \pm .048$
Target-only	$0.360 \pm .052$	$0.336 \pm .051$	$0.304 \pm .040$	$0.331 \pm .053$	$0.301 \pm .041$
Oracle	$0.274 \pm .043$	$0.283 \pm .039$	$0.260 \pm .039$	$0.308 \pm .041$	$0.290 \pm .037$

Table 5: A3, absolute distances. All six strategies, in transformation-statistic space, the instrument this sweep was run with; lower is better.

Paired difference	$K_T = 1$	$K_T = 2$	$K_T = 5$	$K_T = 10$	$K_T = 20$
Transport ahead of reuse z_S <i>half-width</i>	$+1.140^*$ 0.108	$+0.915^*$ 0.105	$+0.526^*$ 0.099	$+0.317^*$ 0.067	$+0.229^*$ 0.059
Transport ahead of target-only <i>half-width</i>	$+0.061^*$ 0.051	$+0.038$ 0.052	$+0.005$ 0.036	$+0.018$ 0.045	$+0.008$ 0.041
Refinement’s gain over transport <i>half-width</i>	$+0.008$ 0.015	$+0.004$ 0.018	$+0.004$ 0.027	$+0.001$ 0.035	-0.029 0.036
Transport + refine behind oracle <i>half-width</i>	$+0.017$ 0.044	$+0.010$ 0.038	$+0.035$ 0.035	$+0.004$ 0.040	$+0.032$ 0.045

Table 6: A3, paired differences. Each row is signed so that a positive value means what the row name says; an asterisk marks an interval excluding zero.

Conclusion. The method beats reusing the source coordinate at every budget, the advantage decaying monotonically by a factor of five and remaining clear of zero throughout. This is the largest effect in the stage. Against target-only modelling the advantage is confined to the smallest budget. **The plan asked for the crossing point to be named: there is no crossing, only a convergence** — the two become indistinguishable from two target images onward, and the method was never shown to fall behind.

Refinement. Refining the transported coordinate on the available target images changes nothing that can be measured here: its gain over transport alone contains zero at every budget, and so does the remaining gap to the oracle. **Refinement neither helps nor hurts once transport has landed close.** Its value is real elsewhere in the same table — reusing the source coordinate improves from 1.439 to 0.521 across the sweep, and that improvement is refinement doing the work — which is the pattern to expect from a procedure that closes distance: useful when the starting coordinate is far away, and invisible when it is not. Per the plan, it therefore stays out of the headline method.

3.4 A4 — Source-abundance sweep

Six source budgets from 16 to 300 images, one target image, 60 episodes each.

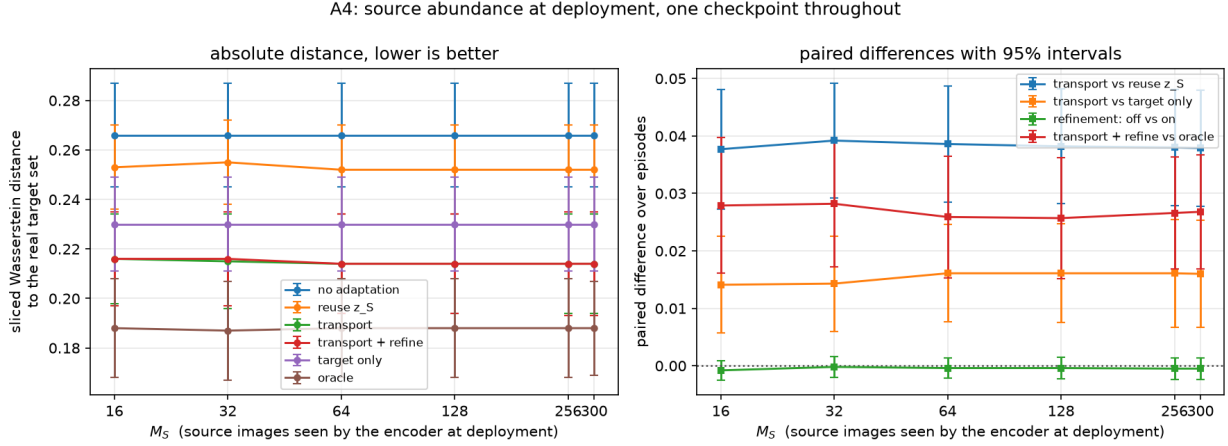


Figure 1: A4. Left: absolute distance per strategy. Right: paired differences with 95% intervals.

Conclusion. Across a nineteenfold range nothing moves: the comparison against reusing the source coordinate shifts by 0.0002, that against target-only modelling by 0.0019, that against the oracle by 0.0011, each smaller than the half-width of its own interval. The plan expected improvement followed by saturation. **What the sweep shows is saturation from the first point measured: sixteen source images already buy everything this transport map extracts from a source set.** The method is therefore cheap on the source side. The sweep varies source evidence available at deployment; the checkpoint was trained once, at $M_S = 64$.

3.5 A5 — Transport-structure ablation

Four maps trained separately under identical conditions, compared episode by episode over 180 episodes each.

Variant	Parameters	$K_T = 1$	$K_T = 5$	$K_T = 20$
Identity	0	$+0.0242 \pm 0.0080^*$	$+0.0256 \pm 0.0089^*$	$+0.0246 \pm 0.0087^*$
Constant relation vector	96	$+0.0082 \pm 0.0051^*$	$+0.0092 \pm 0.0052^*$	$+0.0109 \pm 0.0055^*$
Linear map	352	$+0.0022 \pm 0.0049$	$+0.0004 \pm 0.0048$	$+0.0036 \pm 0.0050$

Table 7: A5. Each variant paired against the residual multilayer perceptron (6824 parameters) episode by episode; positive favours the residual network. The variants differ only in the form of Δ_γ and are otherwise trained identically.

Conclusion. Both reduced maps are beaten at every budget: the identity by $+0.024$ to $+0.026$ and the constant relation vector by $+0.008$ to $+0.011$, every interval clear of zero. **A transport map is necessary, and a per-relation constant offset is not enough** — the prediction has to depend on the source coordinate itself. The linear map, by contrast, is indistinguishable from the residual network at all three budgets with nineteen times fewer parameters, so **the non-linearity is the part that can be dropped**: the architecture can be simplified rather than defended.

One cell qualifies this. At $K_T = 20$ the constant run is also behind on target-only modelling, by $+0.0135 \pm 0.0081$, a strategy that never consults the transport map at all — so at that budget the difference lies somewhere other than in Δ_γ . At $K_T = 1$ and $K_T = 5$ the same comparison contains zero, and the gap is the map’s.

4 STAGE B — GAUSSIAN-MIXTURE POSITIVE CONTROL

Stage B is run entirely on two-dimensional Gaussian mixtures. It touches no image data. That is its purpose: the mixtures are analytically tractable, so the ground truth is known and the mechanism can be checked against it without the instrument ambiguity that attends any measurement on images. Nothing in this section is evidence about CIFAR-100, and the one figure that shows both is labelled accordingly.

Five task families, 32 tasks each, one separately trained checkpoint per family. The families are ordered by how far apart a task’s source and target distributions are placed: the four labelled by a perturbation size are increasingly distant, and the fifth pairs source and target distributions drawn independently. This axis is the relatedness sweep, included so that task distance can be varied deliberately rather than inferred.

Task family	perturb 0.1	perturb 0.25	perturb 0.5	perturb 1.0	unrelated
$L(z_y)$, the task’s own	0.4368	0.4750	0.5234	0.5937	0.4921
$L(\bar{z})$, shared mean	0.5907	0.6178	0.6770	0.7508	0.6453
$L(z_{\text{shuffled}})$, another task’s	0.8183	0.8171	0.8634	0.9345	0.7891
$L(0)$	0.5905	0.6178	0.6729	0.7467	0.6149
Task-specific gain, $L(\bar{z}) - L(z_y)$	+0.1539*	+0.1428*	+0.1536*	+0.1571*	+0.1533*
Profile minimum at	$s = 1$	$s = 1$	$s = 1$	$s = 1$	$s = 1$

Table 8: Stage B, the positive control, on **two-dimensional Gaussian mixtures** — not on images. Five checkpoints, one per task family, each $k = 16$ with a linear decoder trained for 80,000 steps and read at $K_T = 5$ over 32 tasks. Columns are the five task families, ordered left to right by increasing distance between a task’s source and target distributions. The first four rows are the denoising loss under four coordinates: the task’s own, the single mean coordinate shared by all tasks, another task’s, and zero. The fifth row is the reduction in denoising loss obtained by giving a task its *own* coordinate rather than the single coordinate shared by all tasks, paired over the 32 tasks; it is the quantity the whole mechanism is supposed to produce, and it should be positive. The second row reports where the coordinate-loss profile of Figure 2 attains its minimum, with $s = 1$ denoting the task’s own coordinate; a minimum anywhere else would mean the coordinate points in the wrong direction, so this row is the correctness check rather than a measurement of size.

Conclusion. The four-coordinate scan orders itself exactly as the mechanism predicts, in every family: the task’s own coordinate is best, the shared mean and zero are level with each other, and another task’s coordinate is far worse than having none at all. That last point is the informative one. Substituting the wrong task’s coordinate costs roughly 0.23 where removing the coordinate entirely costs nothing, so the coordinate carries task-specific content that actively misleads when it is wrong — which is the signature of a coordinate that is being used rather than absorbed. The profile is minimised at the task’s own coordinate in all five families.

Placed beside the same measurement on images, the well is 26.1% deep on mixtures and 1.6% deep on CIFAR-100, both minimised at the task’s own coordinate. **The mechanism on images is neither absent nor misdirected; it is shallow.** Two observations are unexplained: the relatedness sweep does not move the task-specific term at all although it was built to, and on unrelated tasks the shared mean is *worse* than zero (0.6453 against 0.6149), the only family where that happens.

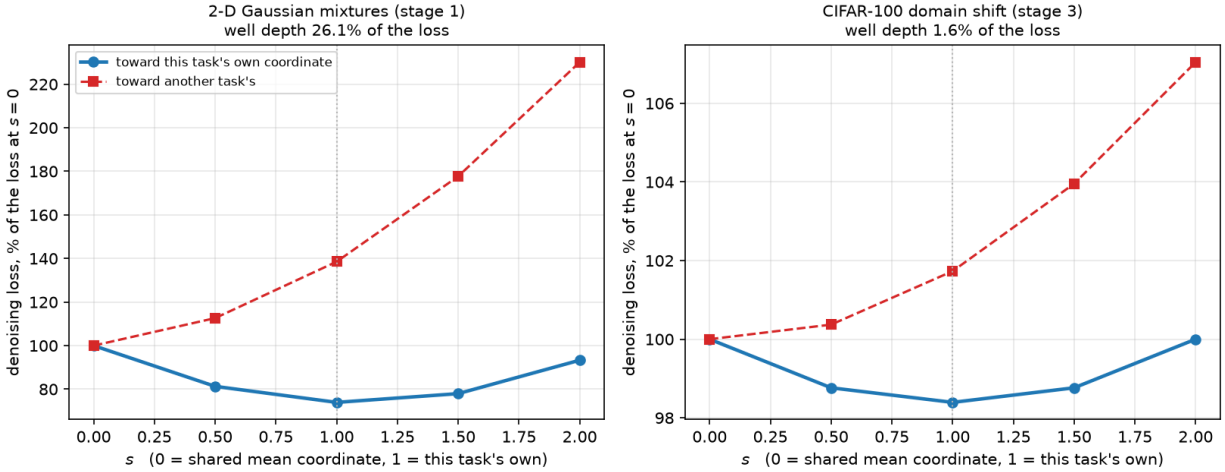


Figure 2: **A cross-stage comparison, not a Stage B measurement.** The left panel is Stage B, on Gaussian mixtures. The right panel is *not*: it is taken from the coordinate controls run on the CIFAR-100 checkpoint in an earlier stage, and is reproduced here only so that the two can be placed on one axis. Both are plotted as a percentage of the loss at the shared mean coordinate, which is what makes them comparable when the absolute loss scales are not. The horizontal axis walks the coordinate from that shared mean at $s = 0$ through the task’s own coordinate at $s = 1$ and out to twice that displacement.

5 STAGE C — DEMOGRAPHIC SCARCITY EXPERIMENT

5.1 C0 — Feasibility check (hard gate)

	Rows	Live images sampled
Archive 1	12,631 (76.2%)	0/30; domain fails to resolve
Archive 2	3,905 (23.6%)	30/30
Conditions above threshold, reachable subset		16
Of those, able to fund a 20-image support set		8
Conditions with the full dataset		83

Table 9: C0. Phototype grouping I–III against IV–VI, chosen from the observed counts.

Conclusion. The annotations are fully available and the images are not. Three quarters of the dataset sits behind a domain that no longer resolves, and recovering it depends on a release the dataset’s authors control. **The reachable subset supports eight usable conditions, which gives a five/two/one task split; one held-out test task cannot carry a paired interval.** It is therefore enough to build C1–C3 on, and not enough for the headline result of C4.

5.2 C1–C5 — Not run

	Item	Reason
C1	Domain module and splits	Not started. Buildable today on the reachable subset; not blocked.
C2	Frozen evaluation predictors	Not started; depends on C1.
C3	Optional sanity check	Not started; depends on C1 and C2.
C4	The scarcity sweep	Blocked by C0. The target-data equivalence statement the plan asks for needs a paired interval on held-out tasks, which eight conditions cannot provide.
C5	Downstream augmentation	Optional and conditional on C4.

Table 10: Stage C beyond the gate. The five items the plan places after C0, none of which was run, with the reason in each case.

6 STAGE D — FEW-SHOT FACE GENERATION

No new runs, as the plan directs. The existing negative result stands, reinterpreted as a task-family mismatch: the standard setup offers a handful of heterogeneous target domains rather than many instances of one reusable relation. A1 supports that reading, since the method’s advantage is concentrated where a reusable relation carries information a single target image does not.

7 DELIVERABLES

#	Deliverable	Status
1	Gaussian-mixture mechanism and calibration	Complete
2	Source-dependence controls	Complete
3	Additive basis against FiLM	Complete; indistinguishable on the headline metric
4	Target-scarcity curve, crossing named	Complete; there is no crossing
5	Source-abundance curve	Complete; saturated from the first point
6	Transport-structure ablation	Complete
7	One demographic scarcity experiment	Not run — blocked by C0
8	Paired 95% intervals for every headline	Complete
9	Downstream augmentation (optional)	Not run — depends on C4
10	Few-shot face stress test (optional)	Not run — excluded by the plan

Table 11: The ten deliverables the plan fixes, and the standing of each at the close of this stage.

8 CONCLUSION

The criterion fixed before the stage began is met: the correct source task together with the learned relation beats both relation-only transfer and target-only modelling at every target budget measured. It is met narrowly at the top of that range — the margin over target-only modelling falls from +0.0167 at one target image to +0.0044 at twenty — and the method is therefore worth most where target data are scarce, which is the setting it was built for. The positive control confirms the mechanism is sound where the ground truth is known and locates the difficulty on images as shallowness rather than absence. Two further results are usefully negative: source evidence saturates below the smallest budget measured, so the method is cheap on the source side, and the non-linear transport map is not distinguishable from a linear one, so the architecture can be simplified.

The architectural claim is the one that did not survive. Against a matched FiLM baseline carrying the coordinate in more than twice as many parameters, the additive basis is indistinguishable at generation on the headline metric, and behind it on several secondary ones; its only clear lead is at the oracle, and the loss-surface explanation advanced for that lead was tested and contradicted. **What the basis can still claim is efficiency: the same generation quality from 55,344 parameters against 118,784.** That is worth reporting, and it is a different paper from the one the stage set out to write.

Two items would change what can be claimed. The unconstrained fine-tuning reference, which is implemented but was not run, would establish whether the remaining gap to the oracle belongs to the task or to the architecture. And the image release behind C0 is the only route to a task family in which a single target example does not already reveal the task — which, on the evidence of A1, is where this method has the most to offer.